

Economics 2220

Principles Of Macroeconomics

Professor Zhigang Feng

Lecture 11:

Aggregate Demand and
Aggregate Supply

► What you will learn in this chapter



- How the aggregate demand curve illustrates the relationship between the aggregate price level and the quantity of aggregate output demanded in the economy
- How the aggregate supply curve illustrates the relationship between the aggregate price level and the quantity of aggregate output supplied in the economy
- Why the aggregate supply curve is different in the short run from the long run
- How the AD-AS model is used to analyze economic fluctuations
- How monetary policy and fiscal policy can stabilize the economy

SHOCKS TO THE SYSTEM



What causes recessions?

Why is inflation high in some recessions and not others?

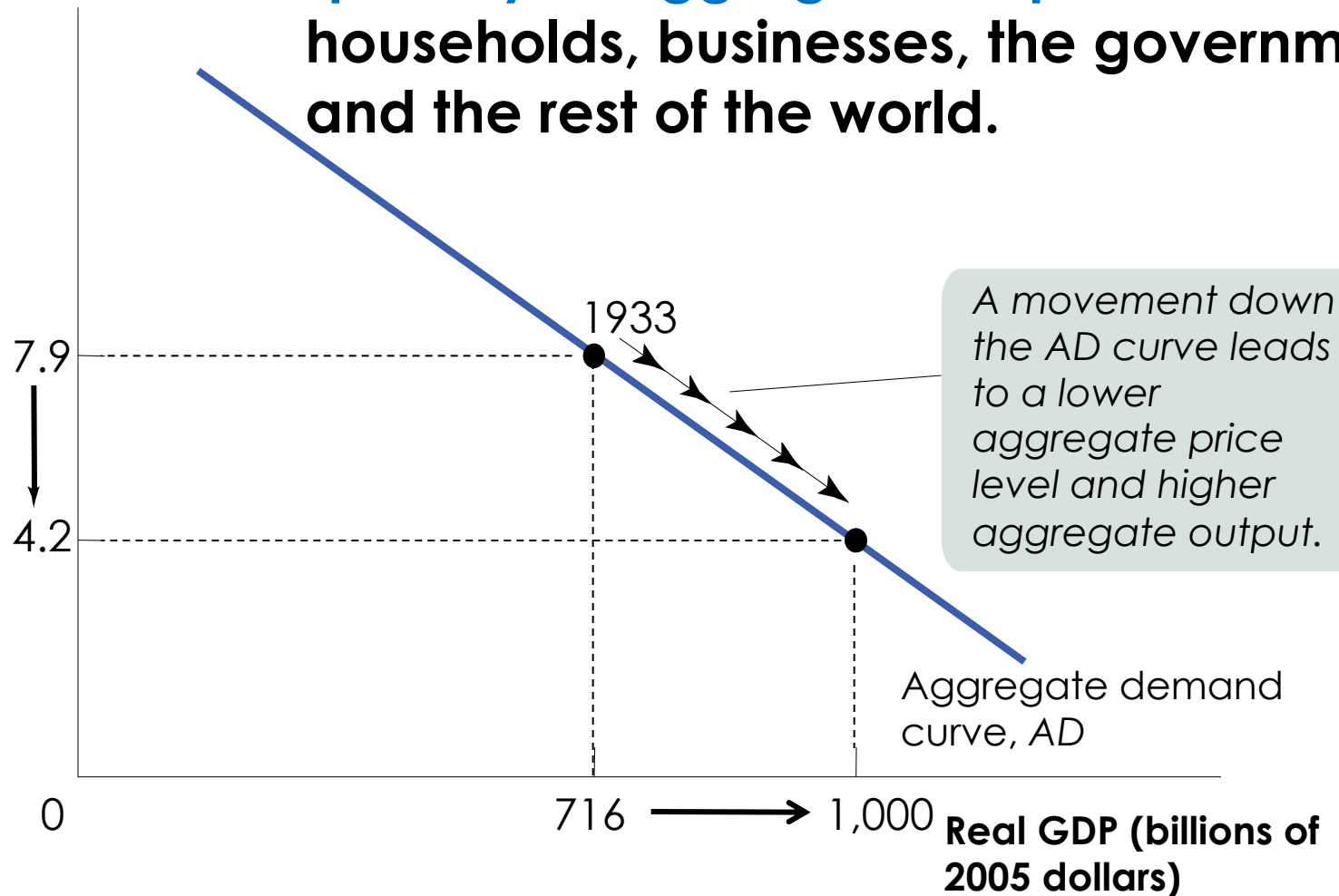
The economy's behavior can be explained, predicted, and potentially manipulated by looking at:

total (aggregate) demand.

total (aggregate) supply.

THE AGGREGATE DEMAND CURVE

Aggregate demand curve: the relationship between the **aggregate price level** and the **quantity of aggregate output** demanded by households, businesses, the government, and the rest of the world.



THE SHAPE OF THE AGGREGATE DEMAND CURVE

Recall: $GDP = C + I + G + X - IM$

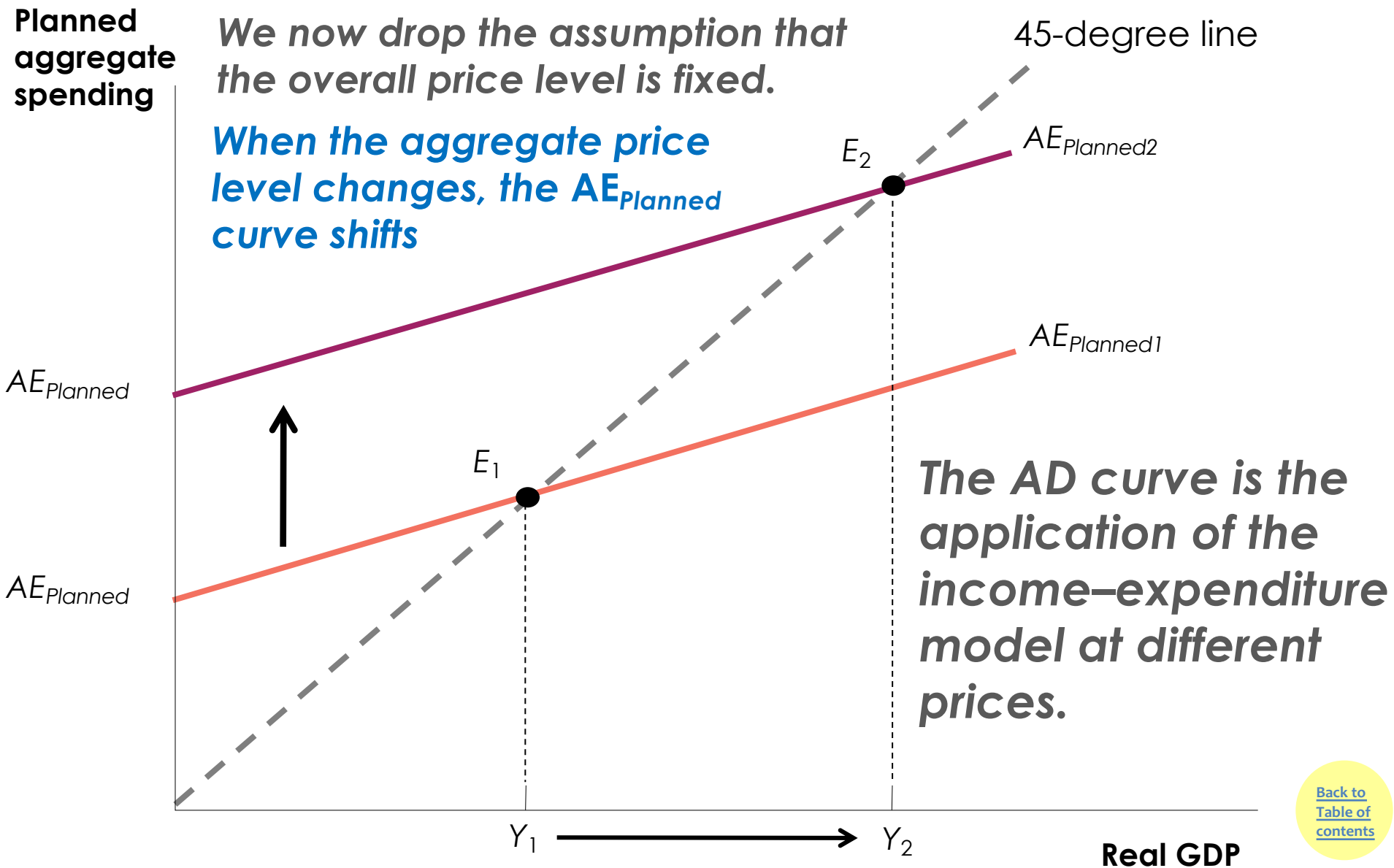
Why does a rise in the aggregate price level *reduce* C , I , and $X - IM$?

(Why does the AD curve slope downward?)

Two reasons:

1. **The wealth effect:** A higher aggregate price level reduces the purchasing power of households' wealth and reduces consumer spending.
2. **The interest rate effect:** A higher aggregate price level makes households hold more money and leads to a rise in interest rates (and a fall in investment spending and consumer spending).

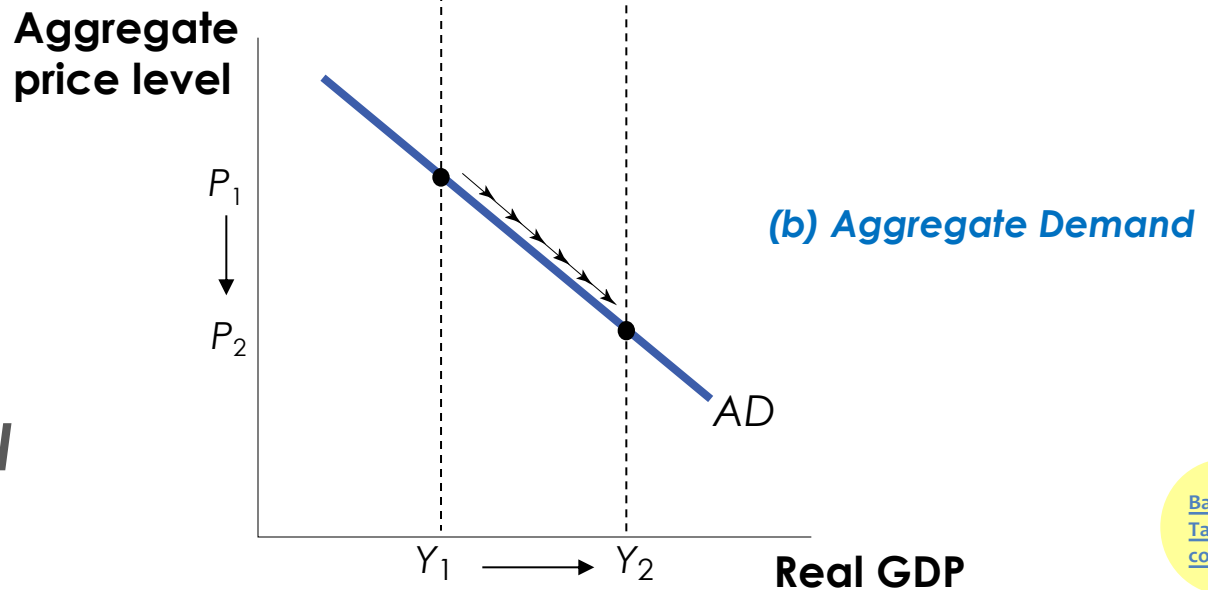
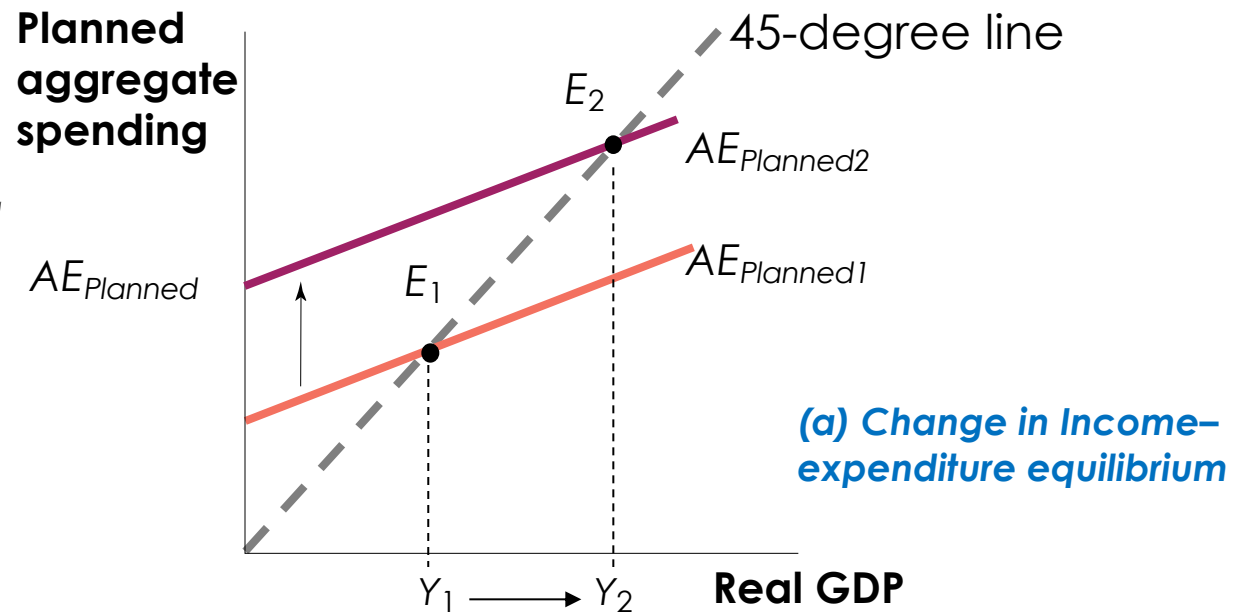
THE AGGREGATE DEMAND CURVE AND THE INCOME-EXPENDITURE MODEL



THE AGGREGATE DEMAND CURVE AND THE INCOME-EXPENDITURE MODEL

If the **price level drops**, **planned spending rises** at all output levels (from the wealth and interest-rate effects).

This leads to a multiplier process that moves the income-expenditure equilibrium from E_1 to E_2 and raises real GDP from Y_1 to Y_2 .



SHIFTS OF THE AGGREGATE DEMAND CURVE

What happens when something changes spending patterns at every price level?

The aggregate demand curve *shifts* because of changes in:

expectations.

wealth.

size of the existing stock of physical capital.

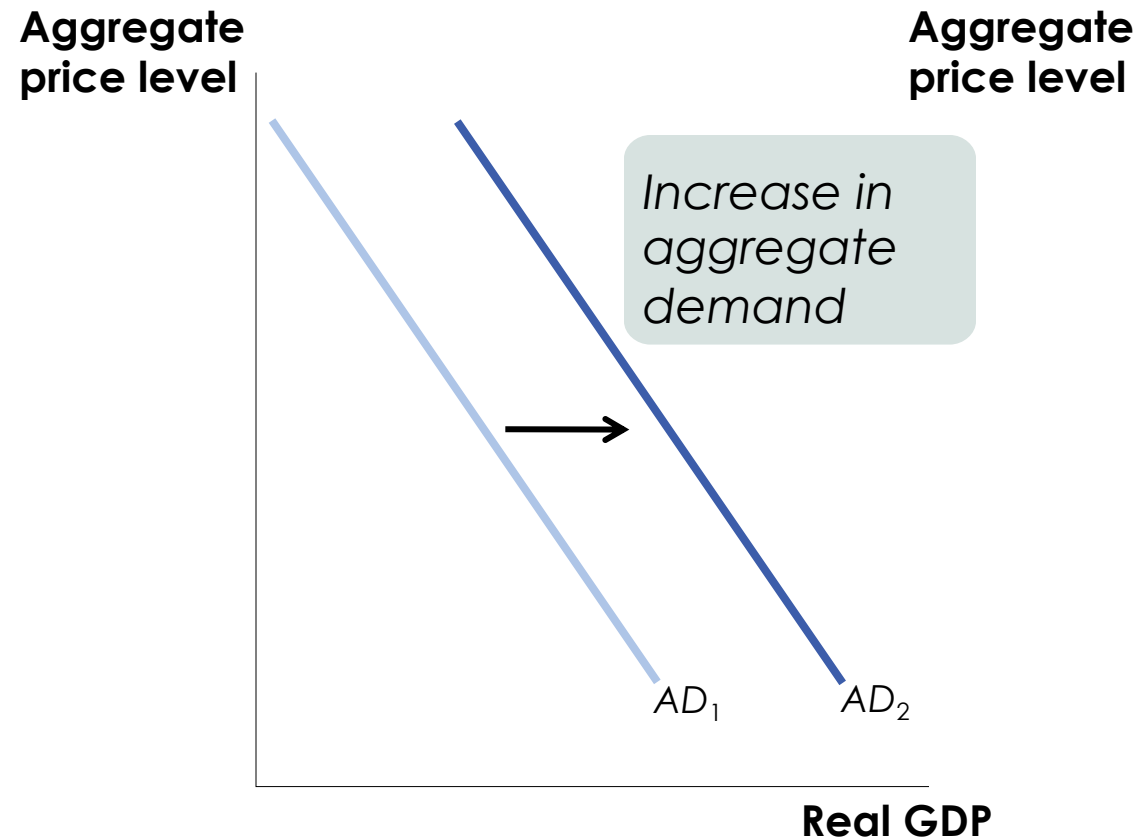
government policies.

Fiscal policy

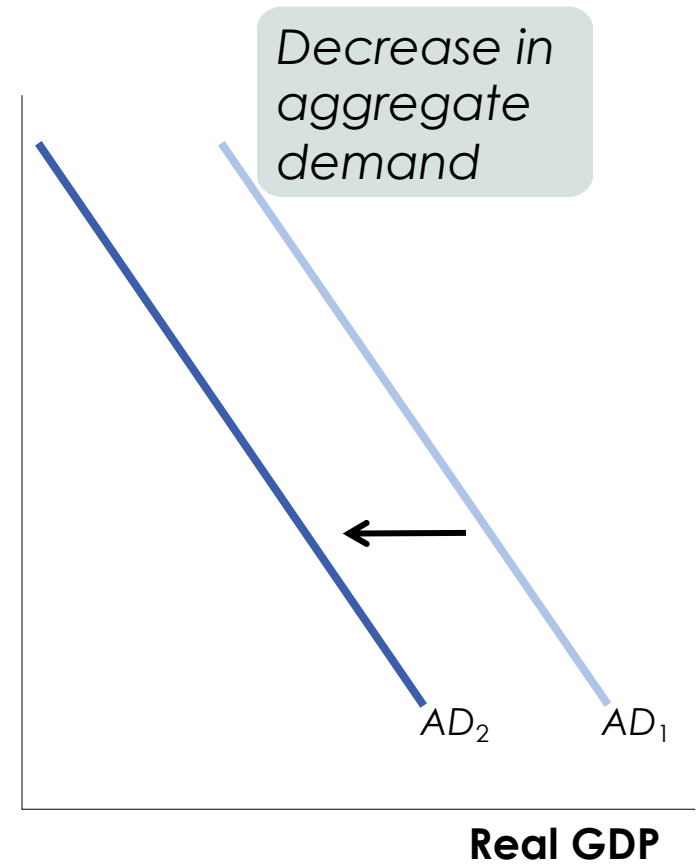
Monetary policy

SHIFTS OF THE AGGREGATE DEMAND CURVE

(a) Rightward Shift



(b) Leftward Shift



CHANGES IN WEALTH: A MOVEMENT ALONG VS. A SHIFT OF THE AGGREGATE DEMAND CURVE

Q: Which is it: Does a change in wealth move us along the AD curve (wealth effect) or shift it?

A: it depends on the **source** of the change in wealth.

If it's a **change in price level** that affects our wealth, it's a **movement along the AD**.

Example: Rapid inflation shrinks our wealth.

If it's a **change in something else** that affects our **wealth**, it's a **shift in the AD**.

Example: The housing market crashes.



LEARN BY DOING: PRACTICE QUESTION

If the economy is in equilibrium and the real estate market collapses, what will likely happen?

- a) The *AD* curve will shift rightward.**
- b) The *AD* curve will shift leftward.**
- c) The *SRAS* curve will shift rightward.**
- d) The *SRAS* curve will shift leftward.**

MOVING ALONG THE AGGREGATE DEMAND CURVE, 1979-1980

Q: How do distinguish between shifts of the AD curve and movements along it?

A: Sometimes it's hard... except in 1979

Prices rose 14.8% while consumer and investment spending dropped.

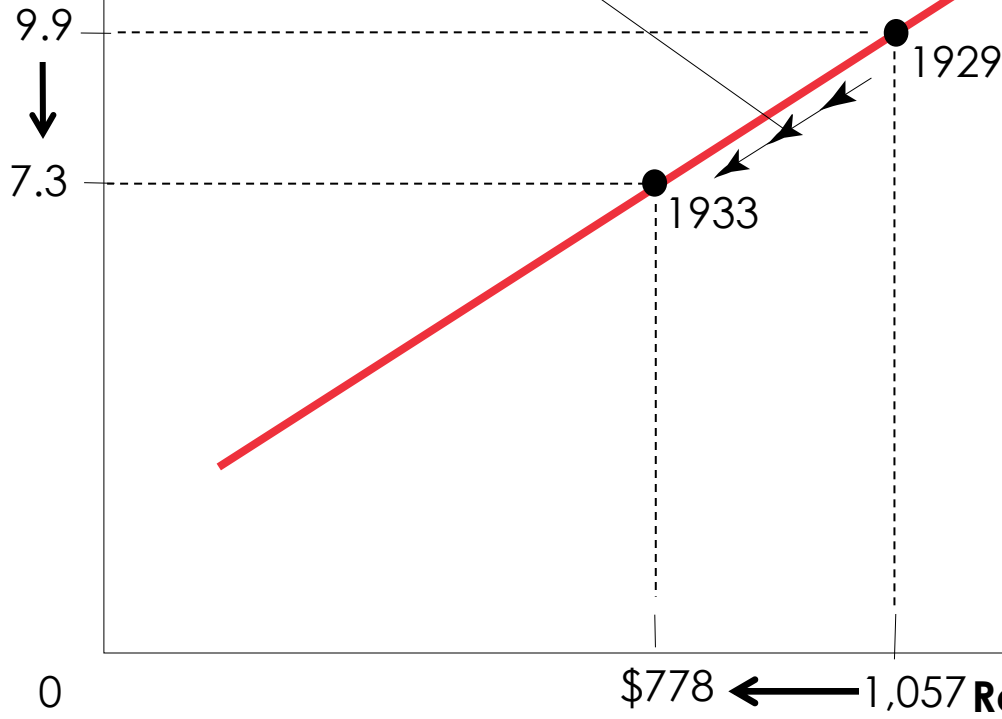
The culprit? A leftward shift in the short-run aggregate supply curve along the AD curve.

THE SHORT-RUN AGGREGATE SUPPLY CURVE

Aggregate price level (GDP deflator, 2005 = 100)

A movement down the SRAS curve leads to deflation and lower aggregate output.

Short-run aggregate supply curve, SRAS



Aggregate supply curve shows the relationship between the **aggregate price level** and the **quantity of aggregate output** in the economy.

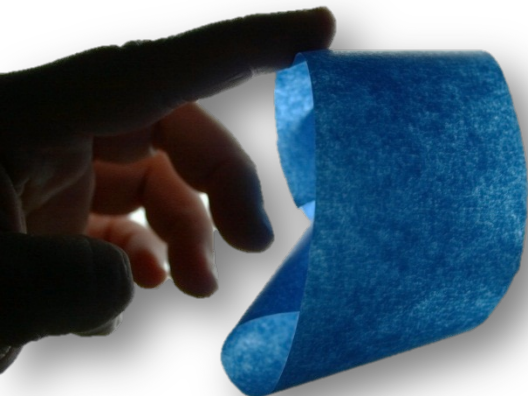
THE SHORT-RUN AGGREGATE SUPPLY CURVE

Why does the *SRAS* curve slope upward?

Because nominal wages are sticky in the short run

Nominal wage: the dollar amount of the wage paid.

Sticky wages: nominal wages that are slow to fall even in the face of high unemployment and slow to rise even in the face of labor shortages.



THE SHORT-RUN AGGREGATE SUPPLY CURVE

How do sticky wages affect SRAS?

Profit per unit = price per unit – production cost per unit.

Nominal wages are often determined by contracts that were signed some time ago.

Even when there are no formal contracts, there are often informal agreements between management and workers.

A higher aggregate price level leads to higher profits and increased aggregate output in the short run.

THE SHORT-RUN AGGREGATE SUPPLY CURVE

One more point:

Many firms are in very competitive industries, so they have no control over price.

Some aren't—and can raise price when demand is strong.

(more support for the positive slope of the *SRAS*)



SHIFTS OF THE SHORT-RUN AGGREGATE SUPPLY CURVE

What happens when something changes production levels at every price level?

The *SRAS* curve *shifts* because of changes in:

Commodity prices.

Nominal wages.

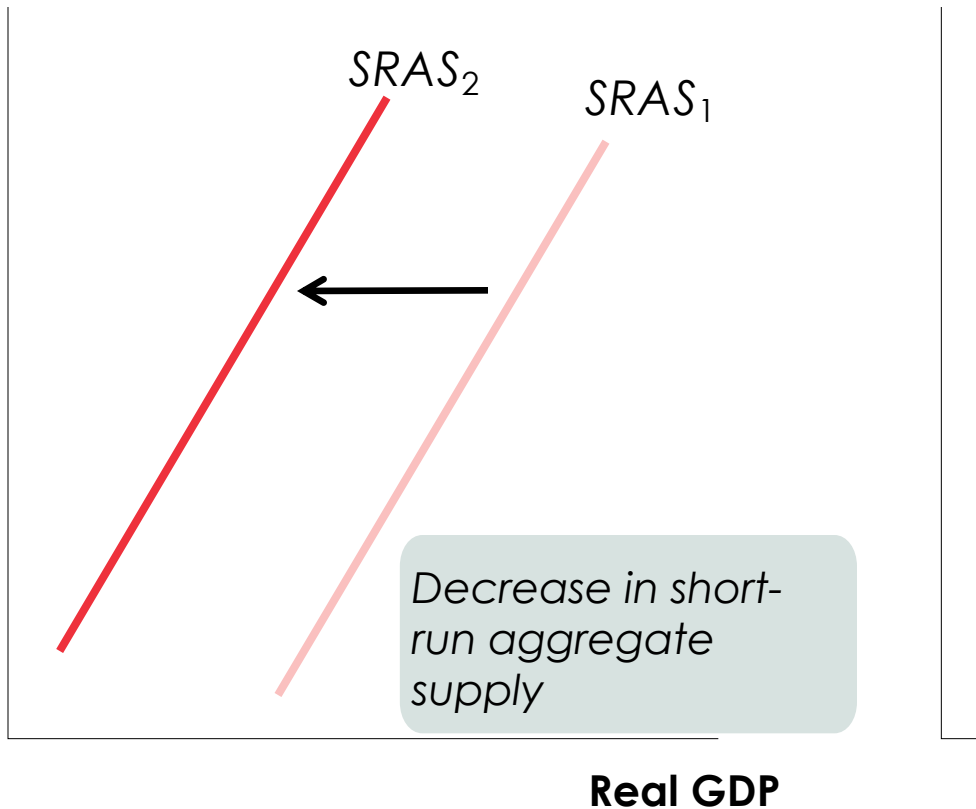
Productivity.

Each of these factors changes producers' profits and therefore shifts the *SRAS*.

SHIFTS OF THE SHORT-RUN AGGREGATE SUPPLY CURVE

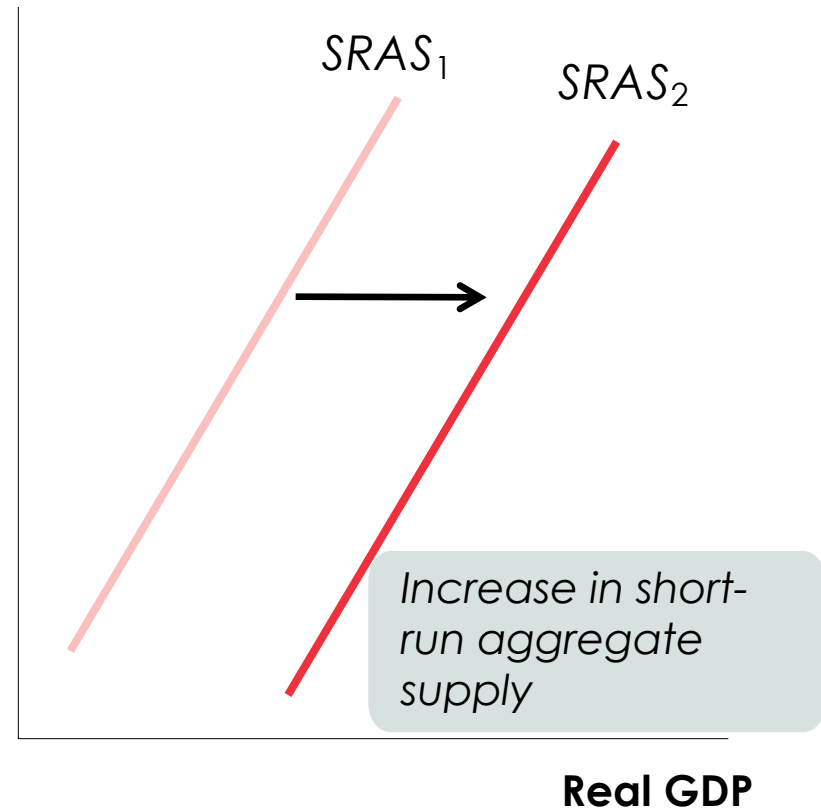
(a) Leftward Shift

Aggregate
price level



(b) Rightward Shift

Aggregate
price level



SHIFTS OF THE SHORT-RUN AGGREGATE SUPPLY CURVE

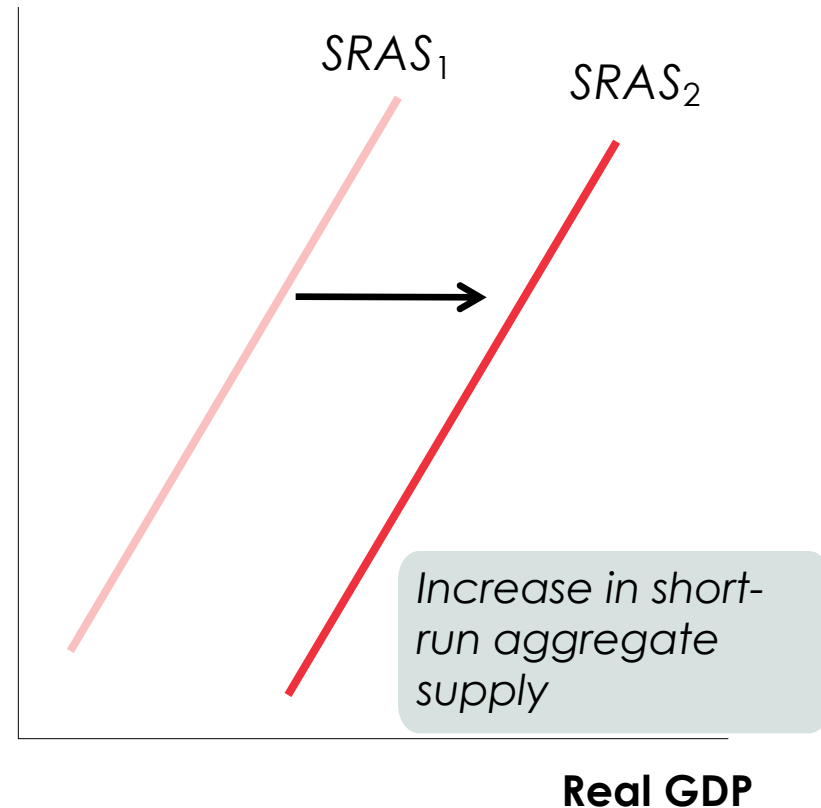
The *SRAS* curve shifts **RIGHT** because of a(n):

- decrease in commodity prices.
- decrease in nominal wages.
- increase in productivity .

Each of these factors increases producers' profits and shifts the *SRAS* right.

Aggregate
price
level

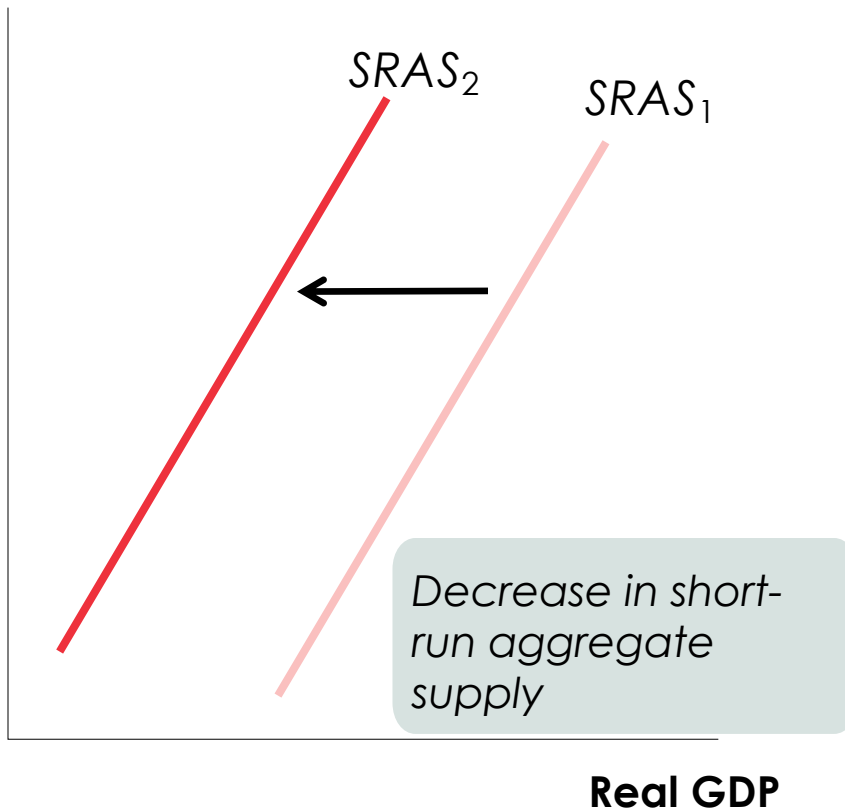
(b) Rightward Shift



SHIFTS OF THE SHORT-RUN AGGREGATE SUPPLY CURVE

(a) Leftward Shift

Aggregate
price level



The $SRAS$ curve shifts **LEFT** because of a(n):

- **increase in commodity prices.**
- **increase in nominal wages.**
- **decline in productivity.**

Each of these factors reduces producers' profits and shifts the $SRAS$ left.

LEARN BY DOING: PRACTICE QUESTION

The short-run aggregate supply curve will shift to the right:

- a) when input costs rise.**
- b) when taxes rise.**
- c) when interest rates rise.**
- d) when productivity rises.**

LONG-RUN AGGREGATE SUPPLY CURVE

The *long-run aggregate supply curve* shows the relationship between the **aggregate price level** and the **quantity of aggregate output supplied** that would exist **if all prices**, including nominal wages, **were fully flexible**.

The long run is the time it takes for all prices (including nominal wages) to adjust.

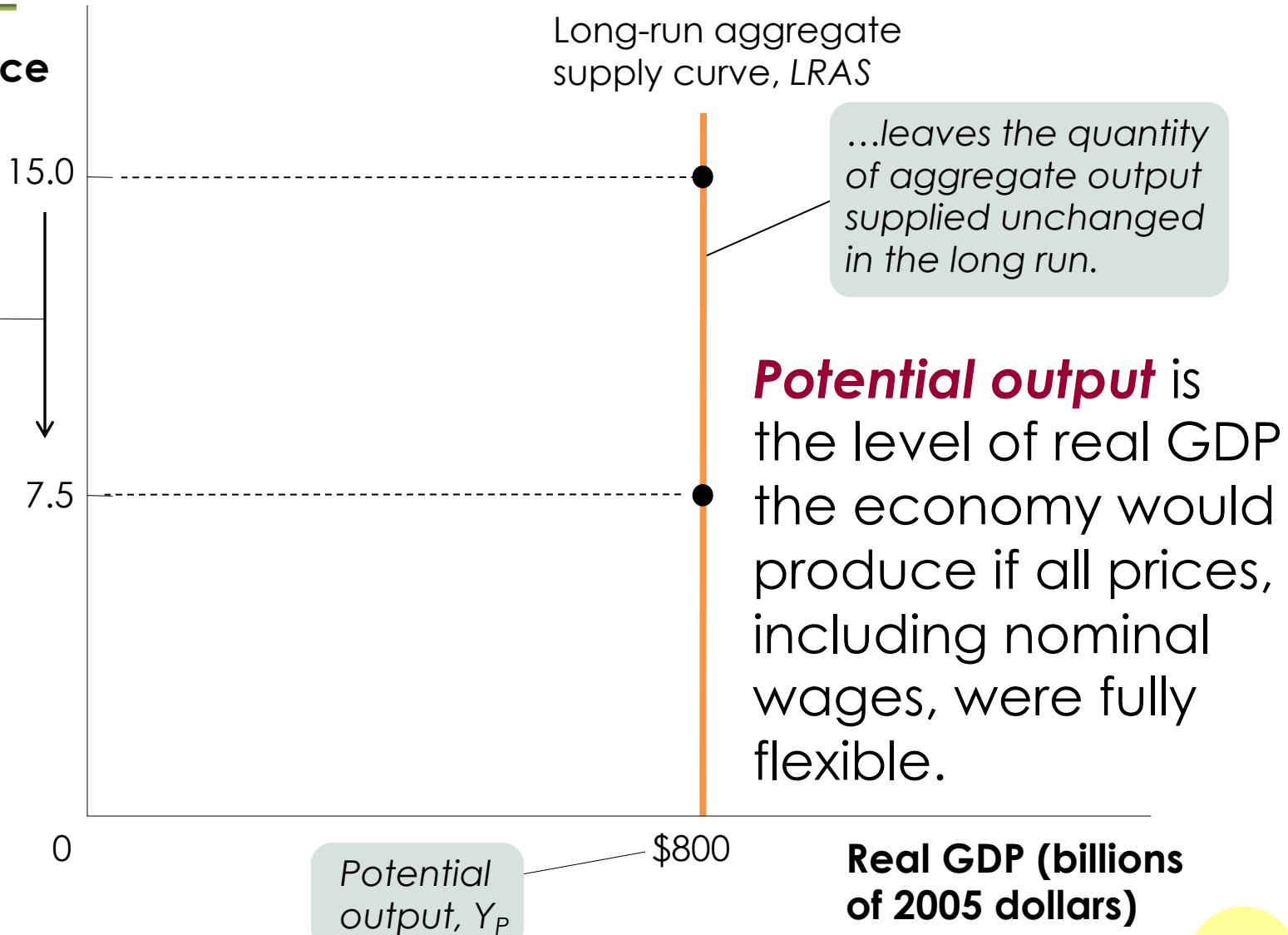
Now there is **no change to profits** SIMPLY BECAUSE PRICES HAVE CHANGED.



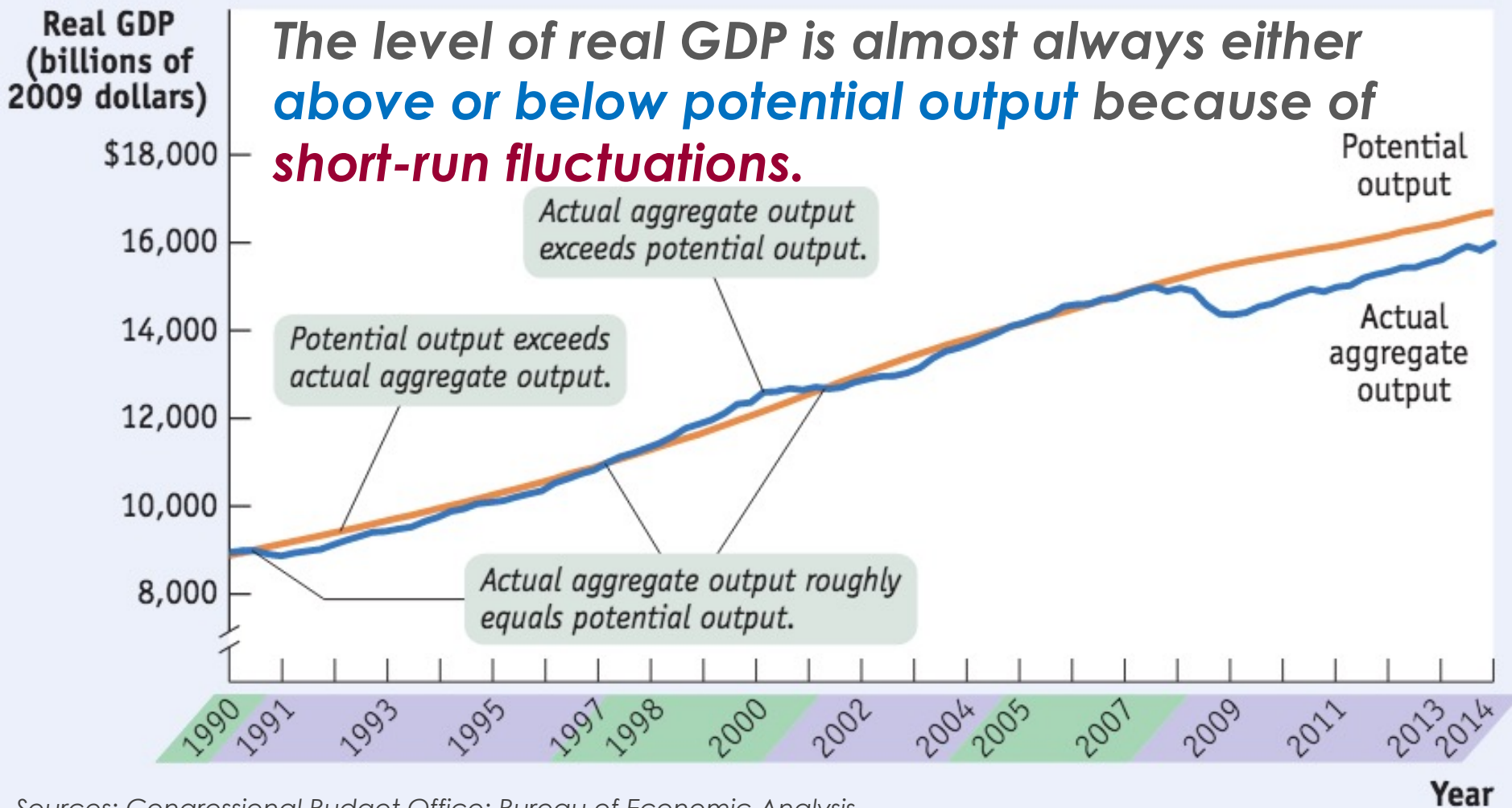
LONG-RUN AGGREGATE SUPPLY CURVE

Aggregate price level (GDP deflator, 2005 = 100)

A fall in the aggregate price level...



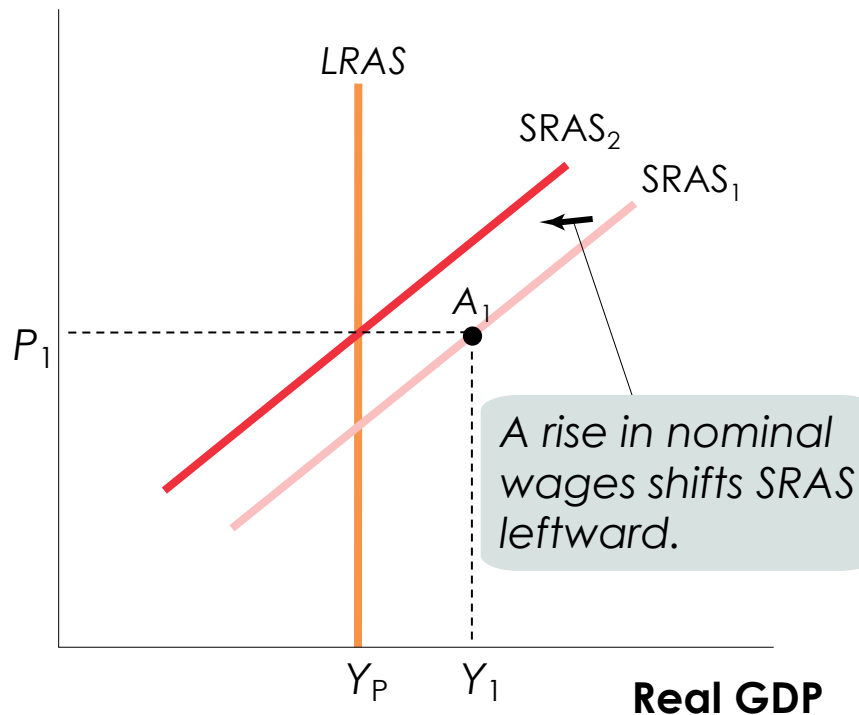
ACTUAL AND POTENTIAL OUTPUT, 1990–2014



Sources: Congressional Budget Office; Bureau of Economic Analysis.

FROM THE SHORT RUN TO THE LONG RUN

Aggregate
price level

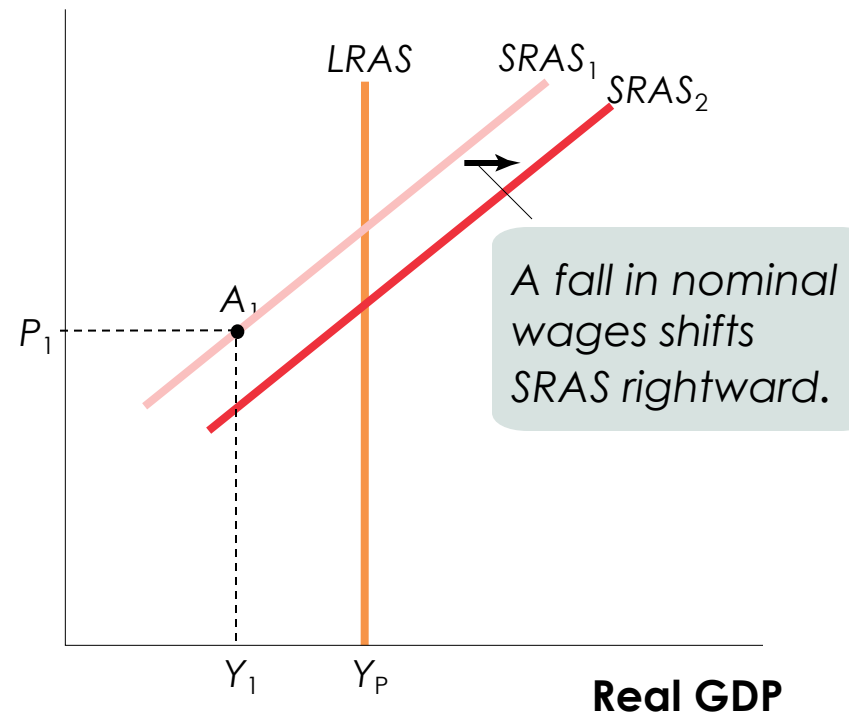


If the economy finds itself at P_1 on $SRAS_1$, aggregate output supplied exceeds potential output (temporarily). **Eventually, low unemployment will cause nominal wages to rise, and SRAS will shift left.**

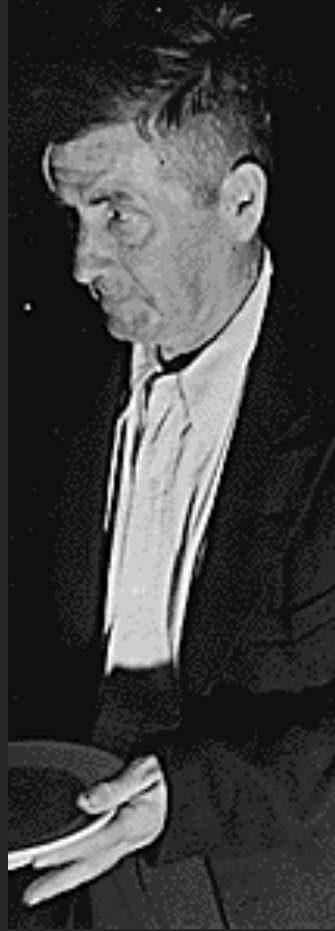
FROM THE SHORT RUN TO THE LONG RUN

If the economy finds itself at P_1 on $SRAS_1$, aggregate output supplied falls short of potential output (temporarily). **Eventually, high unemployment will cause nominal wages to fall, and $SRAS$ will shift right.**

Aggregate
price level

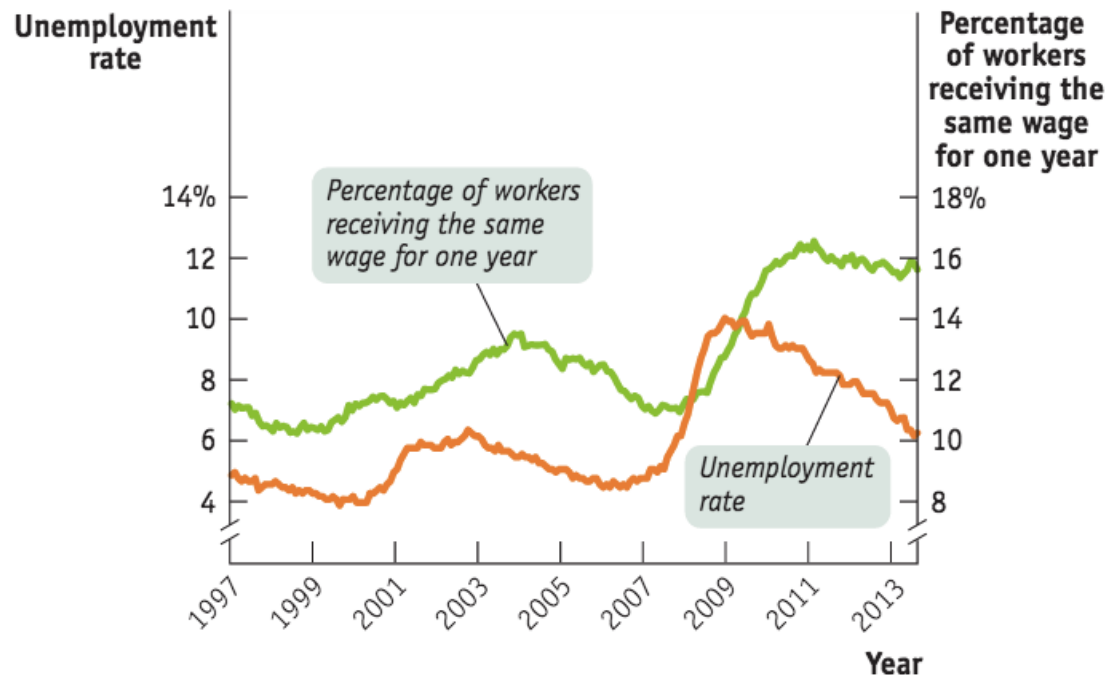


1933: 25% unemployment.
Real GDP had dropped by 27%.
What made it happen?



STICKY WAGES IN THE GREAT RECESSION

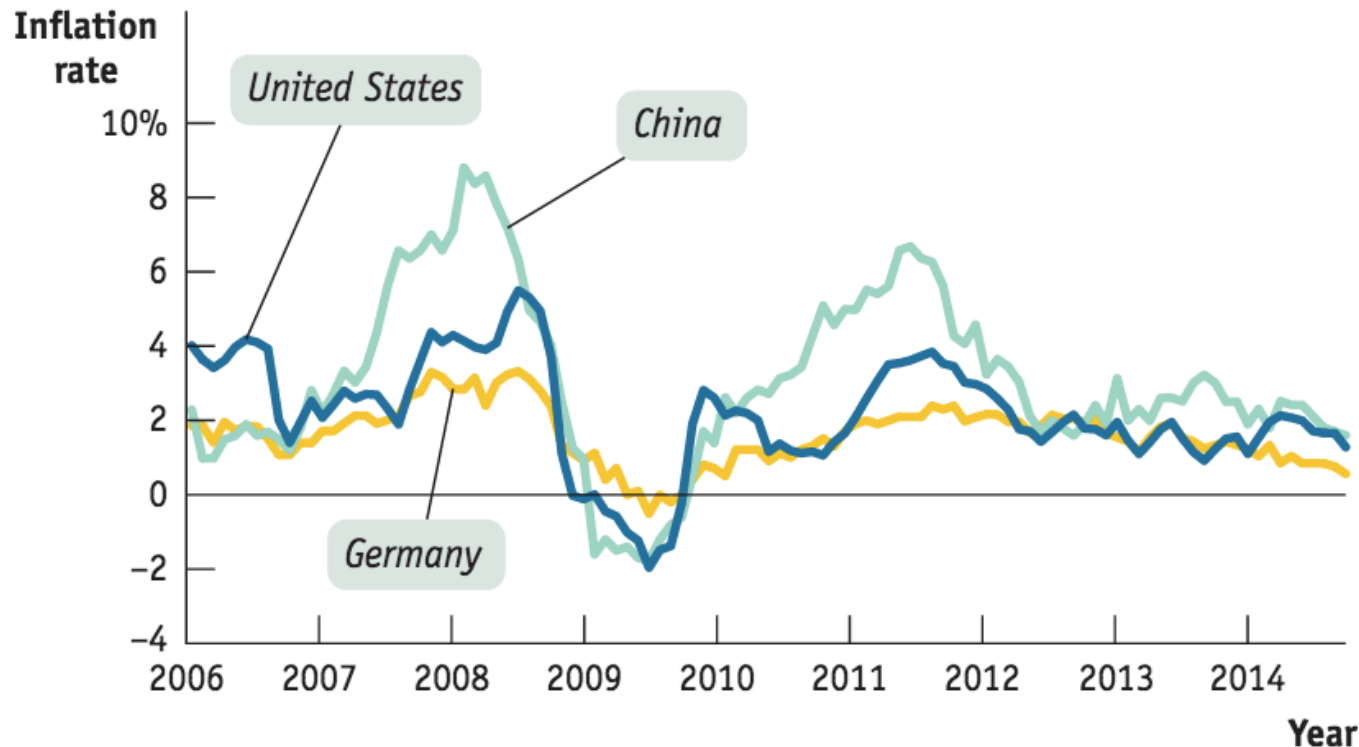
If the theory of sticky wages were true, we'd find evidence that wages fail to fall even during periods of high unemployment. We do.



Sources: Bureau of Labor Statistics; Daly and Hobijn.

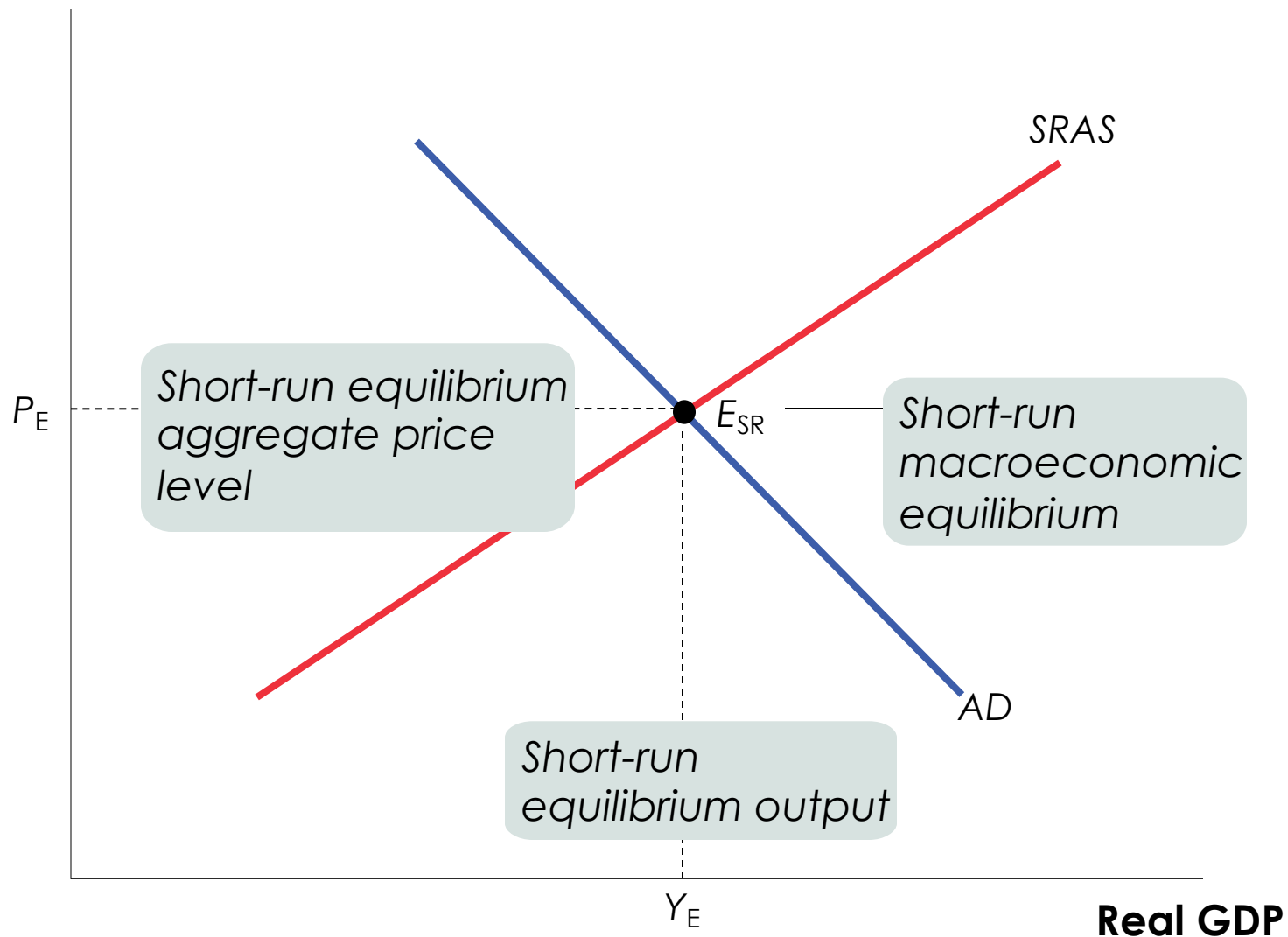
THE AD-AS MODEL

The AD-AS model uses the aggregate supply curve and the aggregate demand curve together to analyze economic fluctuations.



THE AD-AS MODEL

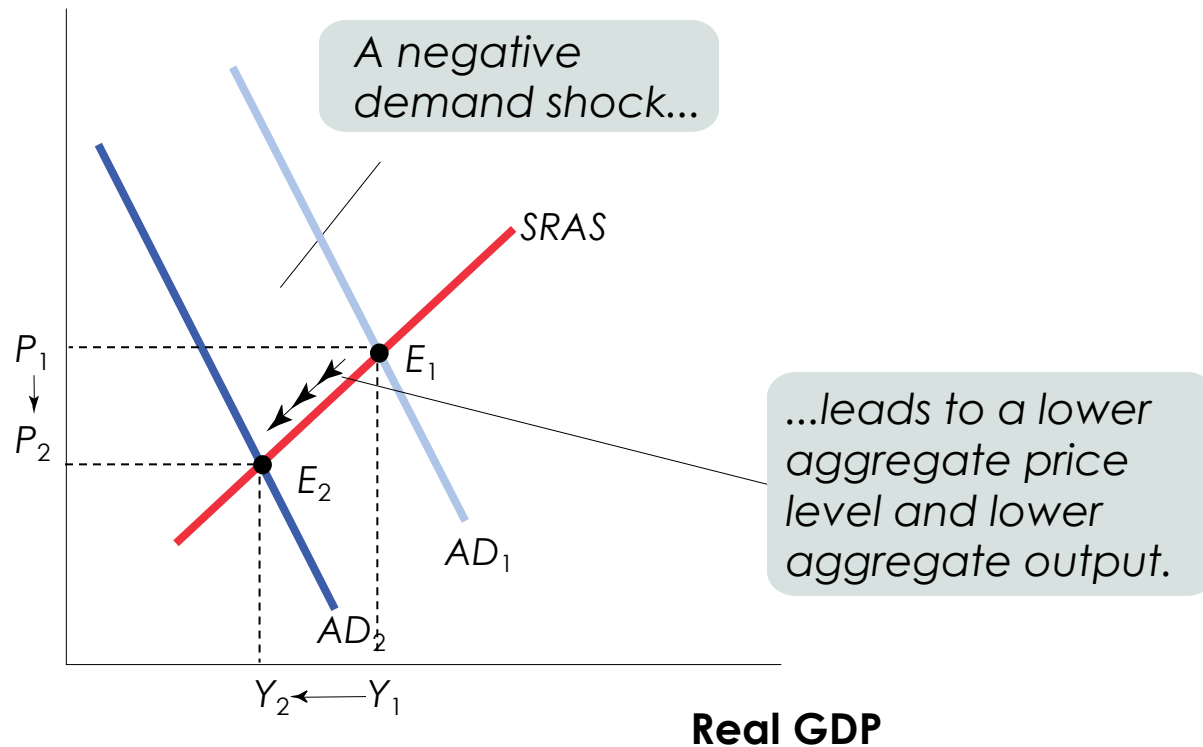
Aggregate price
level



SHIFTS OF AGGREGATE DEMAND: SHORT-RUN EFFECTS

A negative demand shock: Total spending falls.

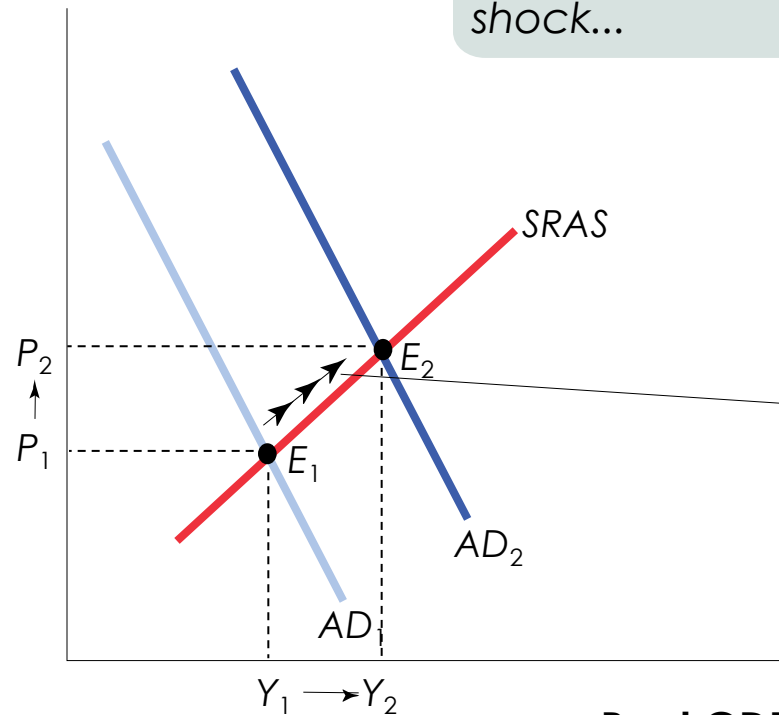
Aggregate price level



SHIFTS OF AGGREGATE DEMAND: SHORT-RUN EFFECTS

A positive demand shock: Total spending rises.

Aggregate price level



...leads to a higher aggregate price level and higher aggregate output.

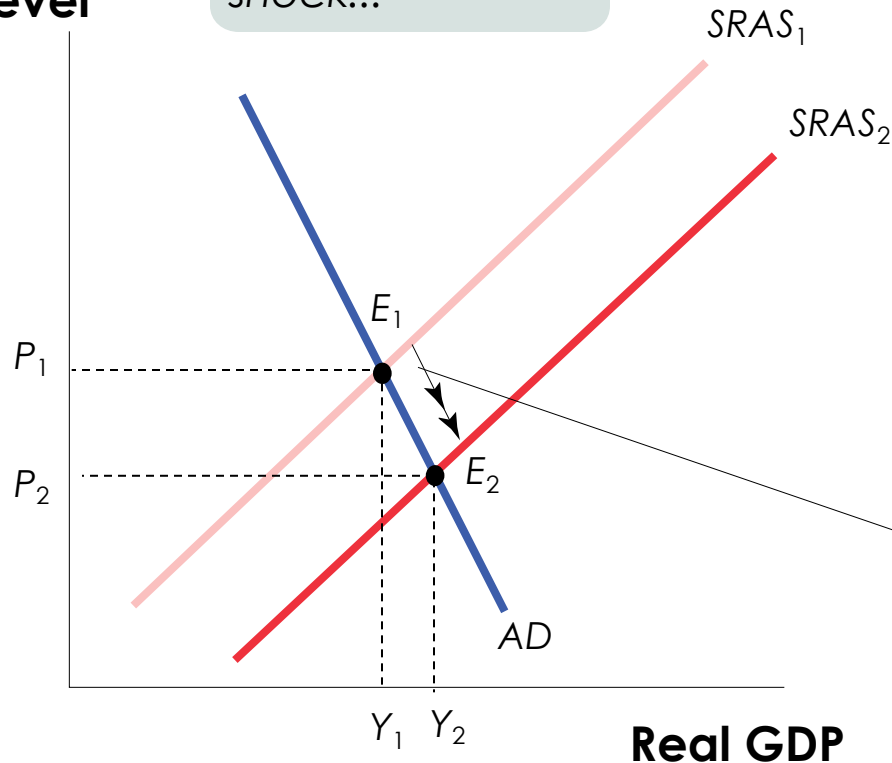
Real GDP

SHIFTS OF THE SRAS CURVE

A positive supply shock: Total production increases at every price level.

**Aggregate
price level**

*A positive supply
shock...*

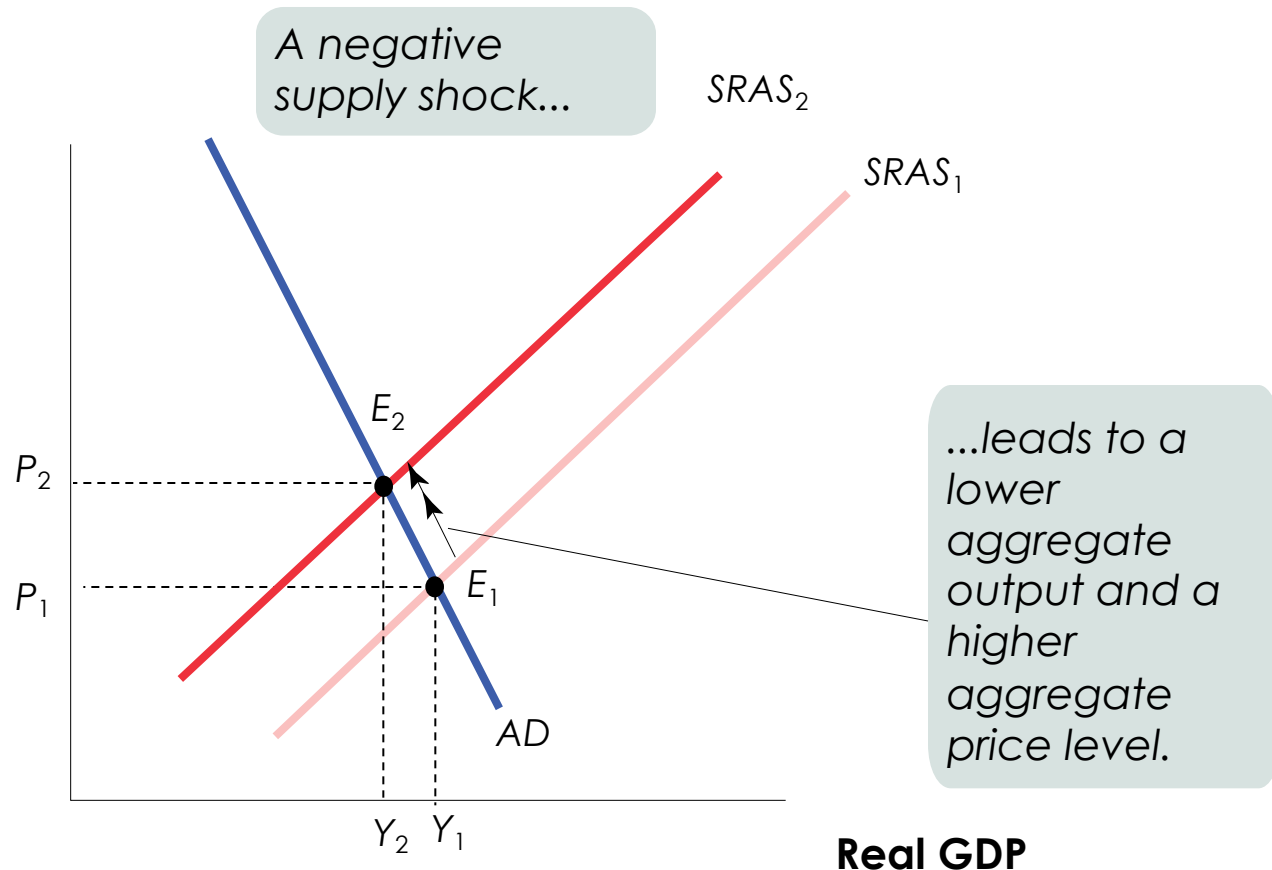


*...leads to higher
aggregate
output, lower
aggregate price
level.*

SHIFTS OF THE SRAS CURVE

A negative supply shock: Total production falls at every price level.

**Aggregate
price level**



NEGATIVE SUPPLY SHOCKS AND STAGFLATION

A diverse group of business professionals in an office setting. In the foreground, a woman with dark hair looks directly at the camera with a serious expression. Next to her, a man with curly hair is looking down at a smartphone. To his right, a woman with glasses is looking down at a blue folder or tablet. In the background, other people are visible, some looking away or talking on mobile phones. The overall atmosphere is one of professional stress or concern.

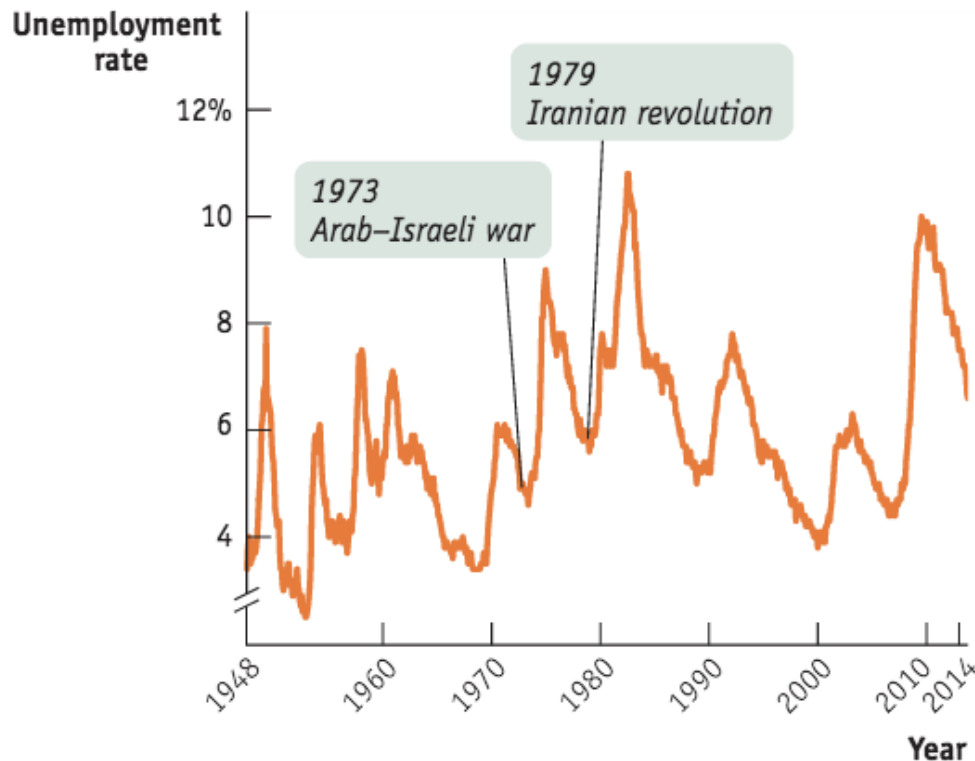
Especially nasty for society is *stagflation*: the combination of inflation and falling aggregate output that comes with a negative supply shock.

ECONOMICS IN ACTION

SUPPLY SHOCKS VS. DEMAND SHOCKS IN ACTION

Which type of recession is more common? Demand shocks are the usual culprit.

Negative supply shocks are rare but nasty.



Source: Bureau of Labor Statistics.

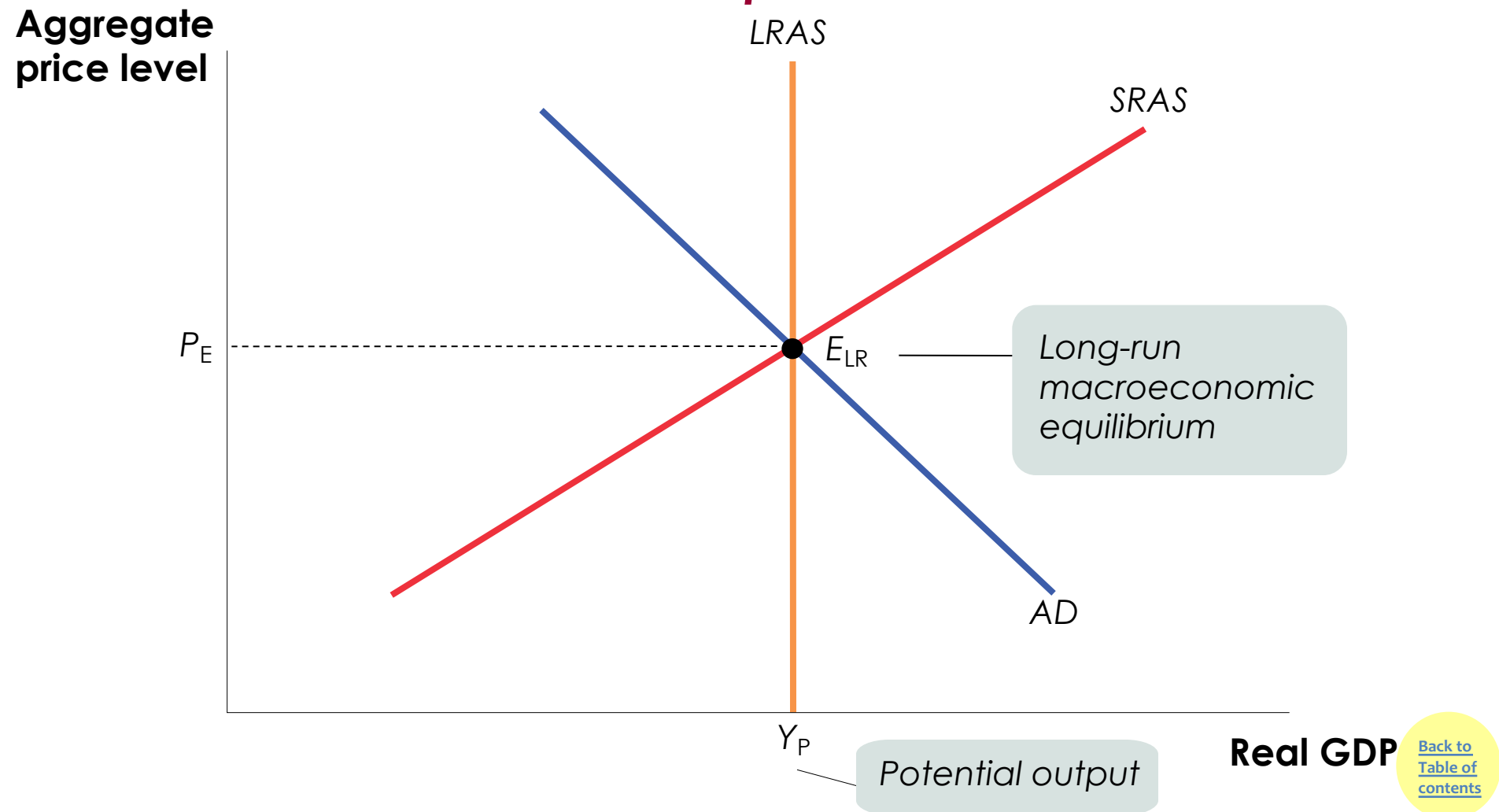
LONG-RUN MACROECONOMIC EQUILIBRIUM

The economy is in ***long-run macroeconomic equilibrium*** when the point of short-run macroeconomic equilibrium is on the long-run aggregate supply curve.



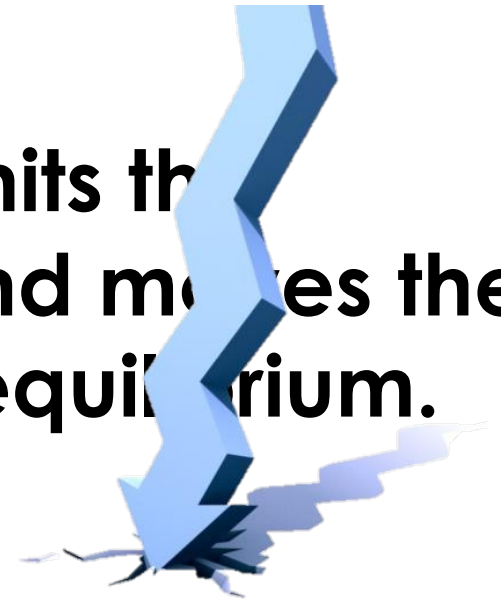
LONG-RUN MACROECONOMIC EQUILIBRIUM

*This economy is in **short run** AND **long-run** macroeconomic equilibrium.*



WHAT HAPPENS WHEN THERE IS A SHOCK?

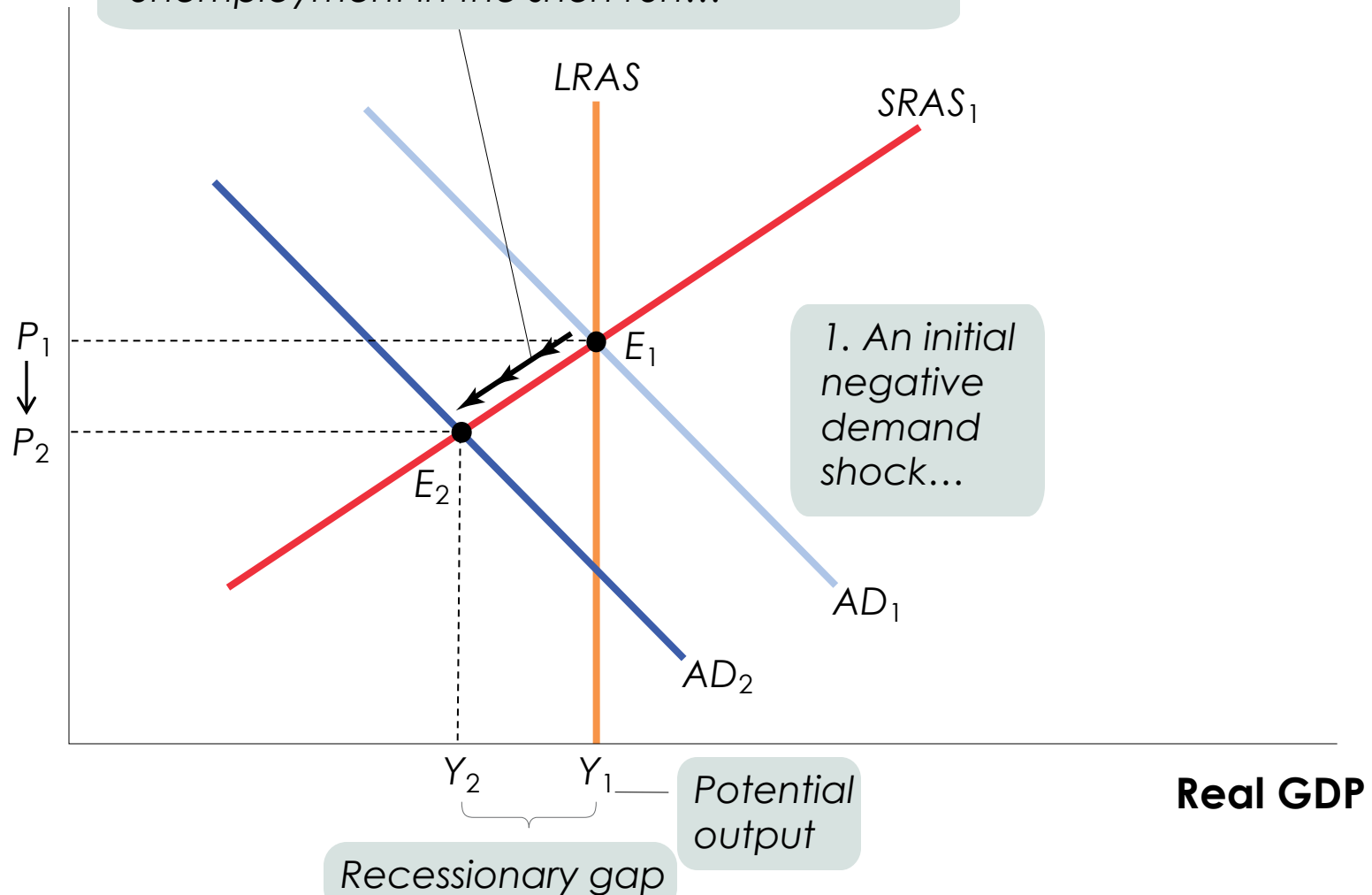
If a demand or supply shock hits the economy, *AD* or *SRAS* shifts and moves the economy to a new short-run equilibrium.



SHORT-RUN EFFECTS OF A NEGATIVE DEMAND SHOCK

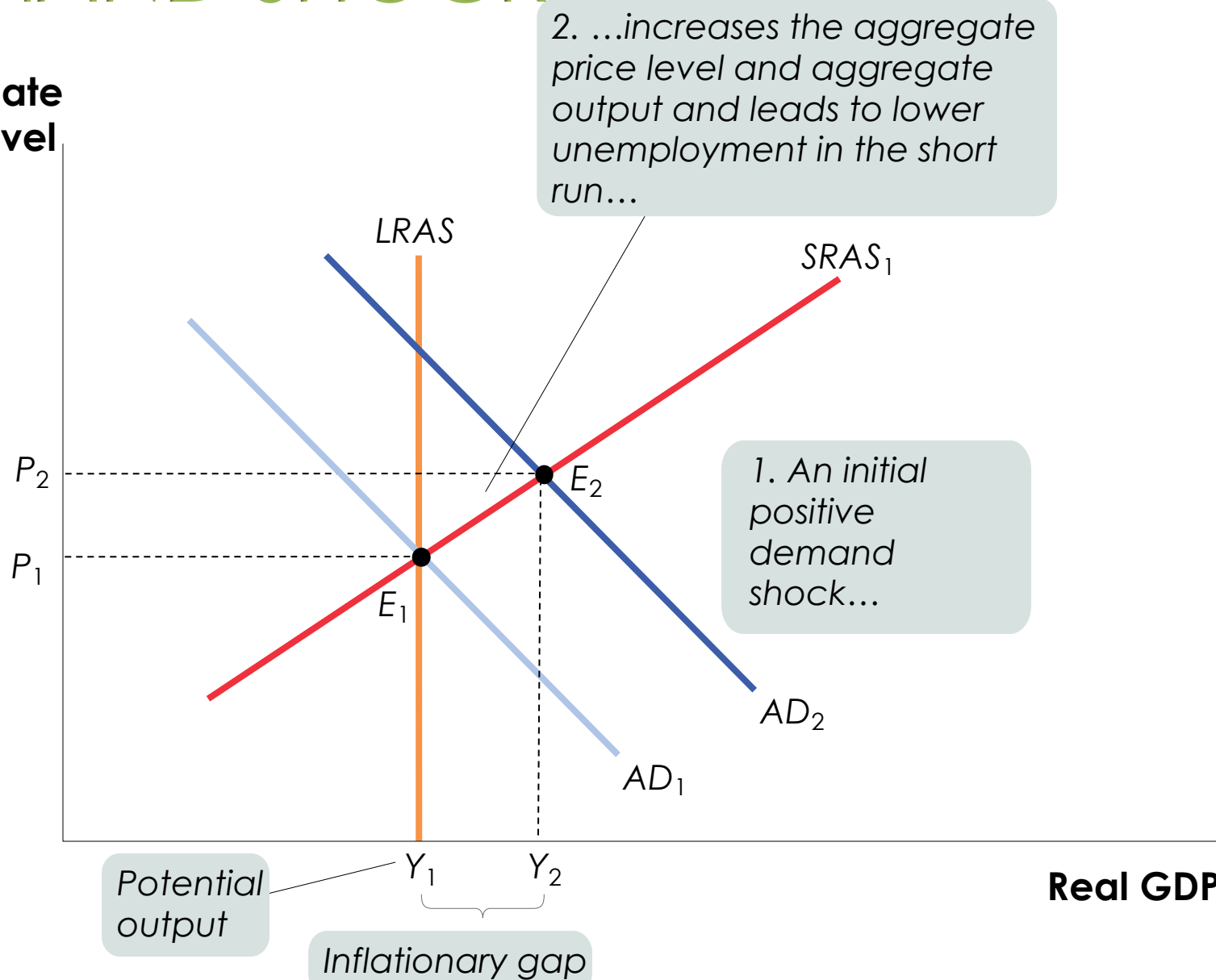
Aggregate price level

2. ...reduces the aggregate price level and aggregate output and leads to higher unemployment in the short run...



SHORT-RUN EFFECTS OF A POSITIVE DEMAND SHOCK

Aggregate price level



GAP RECAP

Recessionary gap: When aggregate output is below potential output

Inflationary gap: When aggregate output is above potential output

Output gap: the % difference between actual aggregate output and potential output

$$\text{Output gap} = \frac{\text{Actual aggregate output} - \text{Potential output}}{\text{Potential output}} \times 100$$

GAP RECAP

What happens next? Are we stuck in a gap forever?

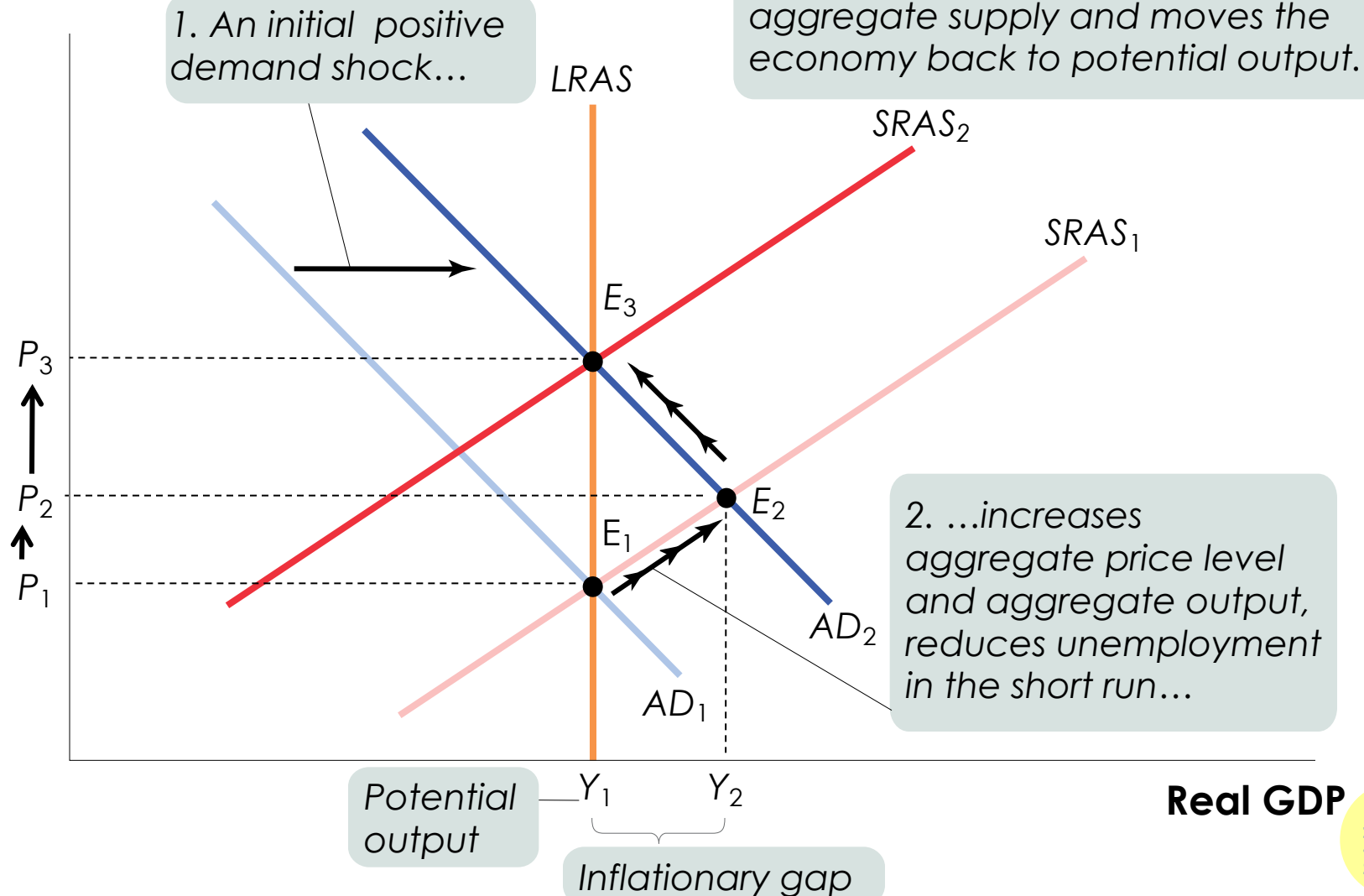
We'll discuss *how policy makers can attempt to move us back to potential output next...*

But the economy may correct itself **in the long run if wages and prices fully adjust.**



SHORT-RUN VERSUS LONG-RUN EFFECTS OF A POSITIVE DEMAND SHOCK

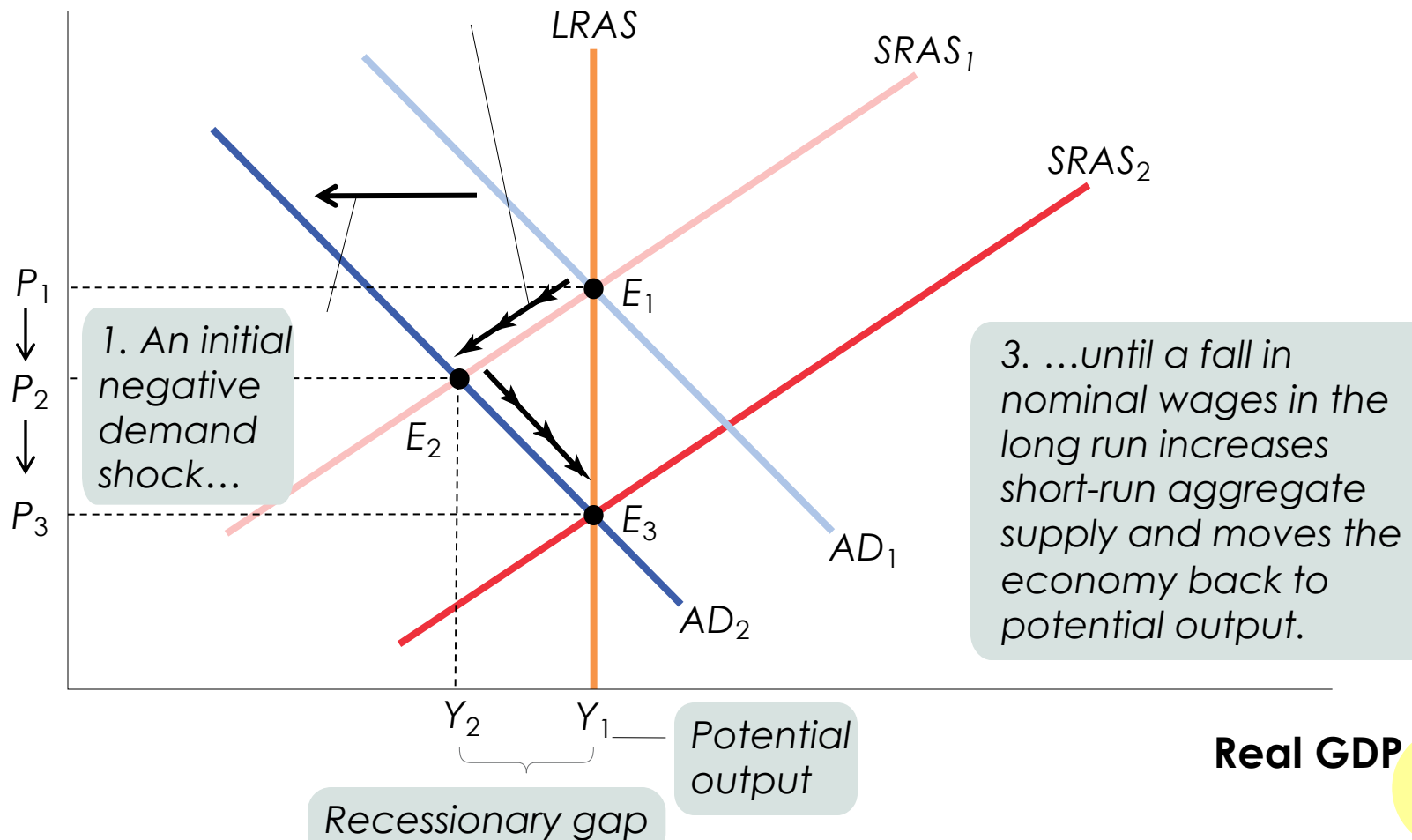
Aggregate price level



SHORT-RUN VERSUS LONG-RUN EFFECTS OF A NEGATIVE DEMAND SHOCK

Aggregate price level

2. ...reduces the aggregate price level and aggregate output and leads to higher unemployment in the short run...



Real GDP

[Back to Table of contents](#)

LEARN BY DOING: DISCUSS

With a partner, answer the following:

The *AD-AS* model is said to have a self-correcting mechanism.

Explain to each other what this means and how this self-correcting mechanism works.

Use a graph to illustrate your answer.

LEARN BY DOING: PRACTICE QUESTION

Suppose an economy is in short-run equilibrium; the level of real GDP is less than potential output. Which of the following statements is true?

- a) In the long run, nominal wages will fall and the *SRAS* curve will shift left, restoring the economy to potential output.
- b) In the long run, nominal wages will fall and the *SRAS* curve will shift right, restoring the economy to potential output.
- c) In the long run, nominal wages will fall and the *AD* curve will shift left, restoring the economy to potential output.
- d) In the long run, nominal wages will fall and the *AD* curve will shift right, restoring the economy to potential output.

LEARN BY DOING: PRACTICE QUESTION

Suppose short-run equilibrium real GDP for an economy is greater than potential output. This implies that:

- a) nominal wages will have to adjust upward as the economy moves from the short run to the long run.**
- b) the level of unemployment is very low.**
- c) jobs are plentiful.**
- d) to reach long-run equilibrium, the *SRAS* curve will shift to the left, resulting in a higher aggregate price level.**
- e) Answers (a), (b), (c), and (d) are all correct.**

LEARN BY DOING: DISCUSS

Use a graph of the $AD-AS$ model to illustrate long-run economic growth in an economy.

Explain your graph and how it illustrates economic growth. Assume in your answer that AD does *not* change over time.

MACROECONOMIC POLICY

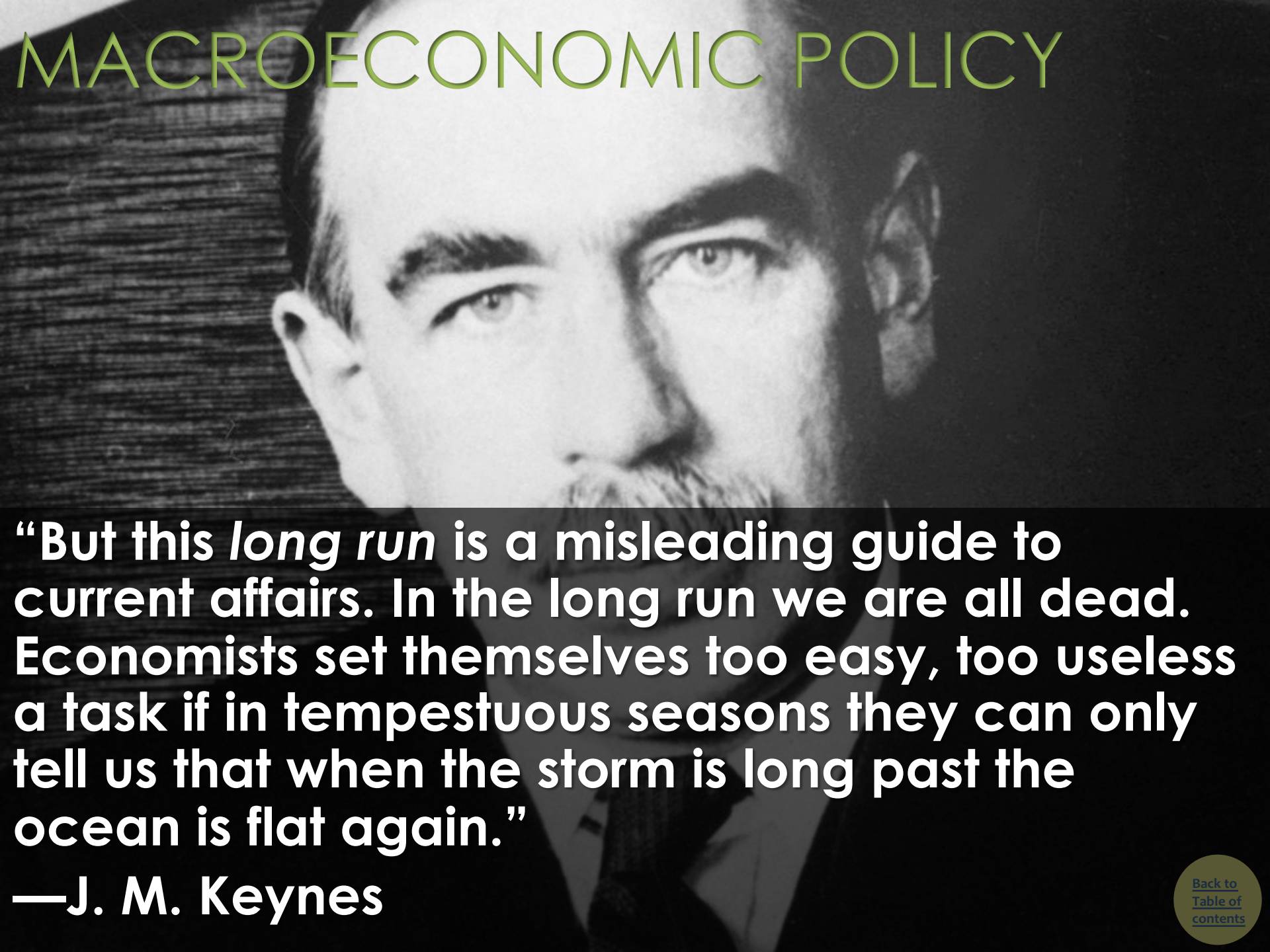
It's true that the economy is self-correcting in the long run.

Most economists think it takes a decade or longer

***Stabilization policy:* the use of government policy to reduce the severity of recessions and rein in excessively strong expansions.**

John Maynard Keynes (1883–1946) created the modern field of macroeconomics.

MACROECONOMIC POLICY



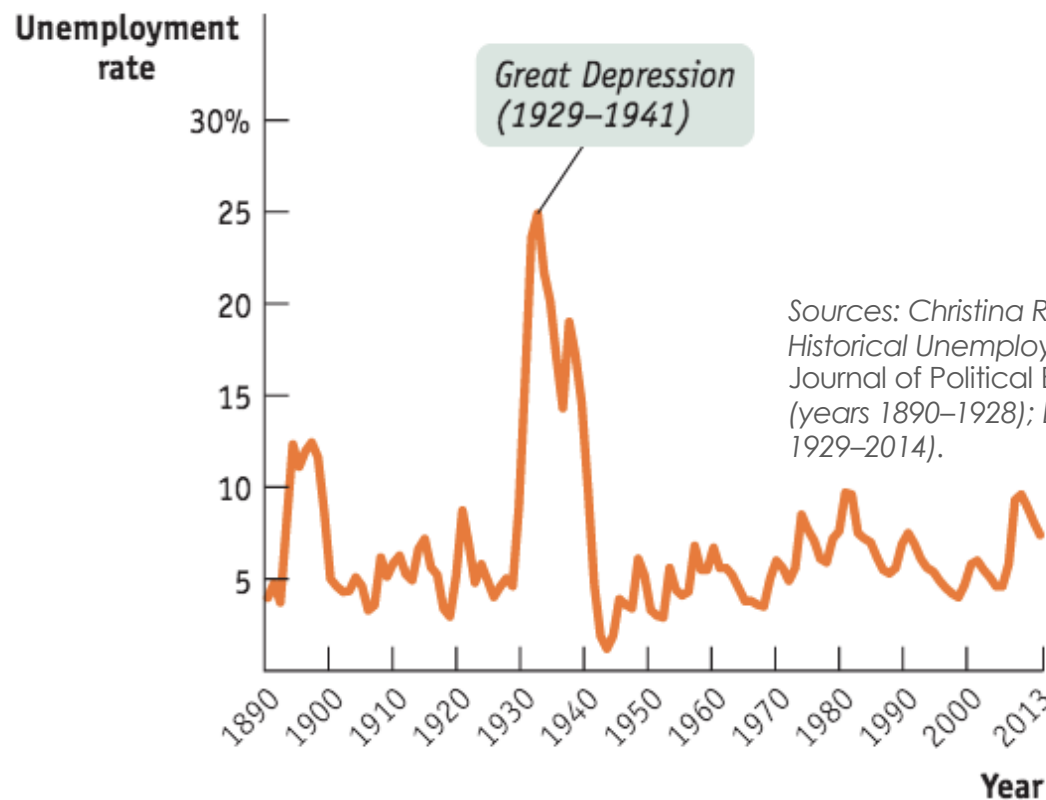
“But this *long run* is a misleading guide to current affairs. In the long run we are all dead. Economists set themselves too easy, too useless a task if in tempestuous seasons they can only tell us that when the storm is long past the ocean is flat again.”

—J. M. Keynes

IS STABILIZATION POLICY STABILIZING?

Governments don't always offset demand shocks; there are costs to intervention.

Unemployment has been more stable since WWII



RESPONDING TO SUPPLY SHOCKS

Negative supply shocks pose a policy dilemma:

To **stabilize aggregate output** requires increasing aggregate demand. This **will lead to inflation**.

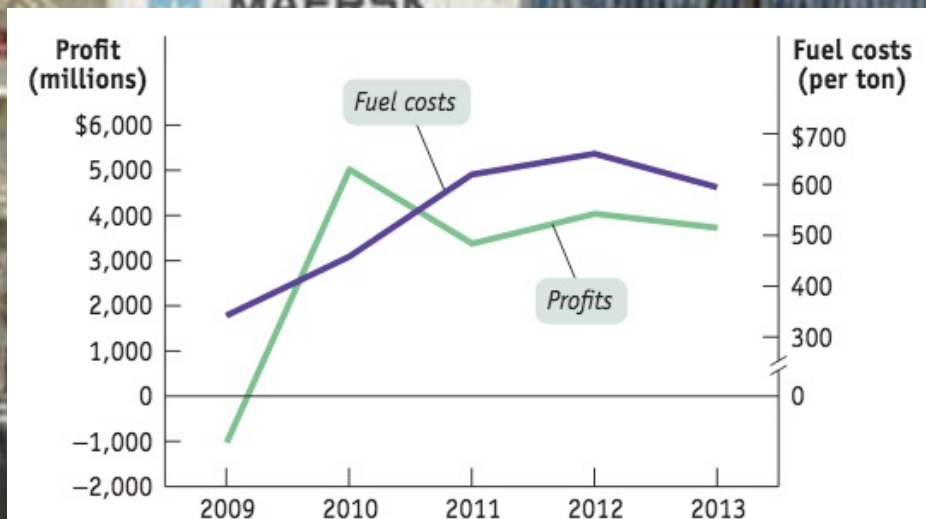
But to **stabilize prices** requires reducing aggregate demand. This **will deepen the output slump**.



Ben Bernanke's job: was harder during the stagflation of 2008 than normal for the chair of the Federal Reserve

LEARN BY DOING: BUSINESS CASE

2011 saw Maersk's profits drop 30% due to rising oil prices... many companies began "slow steaming" to 17 (from 20) knots to save fuel.



Sources: Investor.maersk.com/financialHighlights.cfm; Boston Consulting Group, bcgperspectives.com, "Restoring Profitability to Container Shipping: Charting a New Course," October 10, 2012.